

SRI RAM MANNAM

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EDUCATION

San Jose State University

Master of Science in Software Engineering

San Jose, CA

May 2026

Relevant Coursework: Enterprise Distributed Technologies, Engineering Software Systems, Software Defined Networks, Virtualization Technologies, Machine Learning, Mobile Application Development

Prasad V. Potluri Siddhartha Institute of Technology

Bachelor of Technology in Information Technology

Vijayawada, India

May 2024

SOFTWARE ENGINEERING EXPERIENCE

LessGo – Campus Ridesharing Platform (demo: [demo](#) · code: [repo](#)) *TypeScript, Node.js/Express, Swift/SwiftUI, PostgreSQL, PostGIS, Redis, Python, Docker, GKE*

- Built a full-stack campus ridesharing app with a SwiftUI iOS client talking to 8 Node.js/TypeScript services, covering trip booking, in-app chat, live driver tracking, and Stripe payments so students can actually request and share rides.
- Modeled riders, drivers, routes, and multi-passenger trips in PostgreSQL/PostGIS and wired them into backend APIs, then put everything behind an API gateway with JWT checks and rate limits so mobile calls stayed fast and authenticated.
- Added a 3-stage Python matching pipeline (graph embeddings, PostGIS geo-filtering within 5 km / ±30 minutes, and a cost function across detour, wait time, and social history) to pick drivers that felt fair to both sides instead of just “first available.”

Restaurant Finder – Full Stack Web Application (demo: [demo](#) · code: [repo](#))

React 18, Vite, TailwindCSS,

Python, FastAPI, SQLAlchemy, MySQL, AWS EC2/RDS, GitHub Actions

- Shipped a React 18 + TailwindCSS single-page app backed by a FastAPI/SQLAlchemy REST API so users could search restaurants, read reviews, and manage their own listings behind JWT-protected owner/admin roles.
- Led the front-back glue work for a 4-person Agile team, defining API contracts, wiring ZIP-code search via Google Maps, and building owner/admin dashboards that matched real user flows instead of just CRUD screens.
- Set up GitHub Actions to run unit and integration tests against a SQLite test DB on every push and deploy passing builds to AWS EC2/RDS, so the team could merge UI and backend changes without guessing what was in prod.

DocSense AI – Agentic RAG Document Platform (demo: [demo](#) · code: [repo](#))

Python, Flask, Alpine.js, Qdrant,

FastEmbed, Redis, Supabase, Docker, GitHub Actions

- Built a full-stack PDF “copilot” where users upload documents and chat with them in a simple Alpine.js UI, with the Flask backend returning answers plus page-level citations so people can see exactly where each answer came from.
- Assembled a multi-stage retrieval pipeline (dense vectors via FastEmbed, sparse BM25, RRF fusion, cross-encoder reranking, and a Corrective RAG loop) to handle messy, multi-hop questions instead of just keyword search.
- Indexed both text and vision captions, added a three-tier Redis cache, and enforced per-user isolation with Supabase JWT auth and separate Qdrant collections, then shipped it all as a Docker image to Render with GitHub Actions.

EdgeRunner – Autonomous Trading Platform (demo: [demo](#) · code: [repo](#)) *Python, FastAPI, React 19, TypeScript, TailwindCSS, PostgreSQL, Alpaca API, pytest*

- Built a FastAPI backend and React 19 + TypeScript dashboard exposing 15+ endpoints for positions, approvals, kill-switch controls, and performance charts so humans can see and override what the trading bot is doing in real time.
- Implemented backtested quant strategies behind an 8-check risk gate and idempotent trade execution, tracking Sharpe ratio, max drawdown, and win rate so strategy changes are tied to real portfolio behavior.
- Wrote 241 offline pytest tests around the risk gate, portfolio math, and execution paths, all passing without API keys or network access, which kept refactors from silently changing how trades are placed.

ADDITIONAL EXPERIENCE

Wells Fargo Software Engineering Virtual Experience – Forage (cert: [link](#))

Feb 2026

- Designed a normalized ERD and implemented relational entities in Java using OOP, mirroring accounts, customers, and transactions in code and schema, then published the solution to GitHub.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java, SQL, C++

Frontend: React, React Native, HTML5, CSS3, TailwindCSS

Backend: Node.js/Express, Flask, FastAPI, Spring Boot, RESTful APIs, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Qdrant (vector)

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda), GCP, Docker, Kubernetes, GitHub Actions, CI/CD

AI & Data: OpenAI/LLM APIs, LangChain, FastEmbed, RAG pipelines, scikit-learn

Testing & Quality: pytest, Jest, unit testing, integration testing, code reviews

Other: Git, JIRA, Agile/Scrum, system design, data structures and algorithms

LEADERSHIP EXPERIENCE

SmartInterviews – Student Mentor (cert: [link](#))

May 2023–Mar 2024

- Mentored 100+ students in data structures, algorithms, and full-stack engineering through live sessions and async feedback, including 50+ project/code reviews with concrete comments on API design, React patterns, and Git workflows.
- Wrote and delivered a small curriculum on backend development and frontend integration, turning topics like REST, pagination, and error handling into step-by-step examples that students could build and debug on their own.